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Paleomagnetic Evidence for Post-Jurassic Stability of Southeastern Mexico: Maya Terrane

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The tectonic evolution of southeastern Mexico has been a subject of major controversy, not only in regard to past geometry but also in the timing of proposed geological events as well. For the past 10 years, most, if not all, investigators agree that the Gulf of Mexico Basin was formed by Late Jurassic time and that the Maya Terrane was in its current location prior to the Cretaceous. In order to gain further insight into the drift history of the Maya Terrane we have undertaken a paleomagnetic study of the uppermost Jurassic-Lower Cretaceous (Tithonian-lower Neocomian?) San Ricardo Formation in southeastern Mexico, at 93.7°W, 16.8°N. The sampling site is located east of the Isthmus of Tehuantepec, on the southwest side of the Maya block, at the base of the Yucatan Peninsula. A suite of 133 samples was collected in stratigraphic succession from a 114-m-thick sequence of red shales and sandstones near Cintalapa, Chiapas, Mexico. After progressive thermal demagnetization of all samples at six steps from 350°C to 630°C, 89 samples were selected for final paleopole analysis on the basis of their magnetic stability. Four different polarity intervals were observed, the sequence being from bottom to top: N, R, N, R which assists in the assessment of the reliability of the observations. The mean pole position obtained, 160.0°E, 69.8°N, agrees with the mean pole position of the upper part of the Morrison Formation of Colorado, a unit of virtually identical age. These results indicate that no discernible rotation or displacement of the Maya block has occurred since at least early Neocomian times.

INTRODUCTION

During the last decade an ever increasing number of worldwide paleomagnetic studies on suitable rocks of all ages have provided a better understanding of tectonic movements that have occurred in the Earth's crust. Pole positions for most geologic periods and polar wander curves for some continents are now available, against which new paleomagnetic data from potentially structurally deformed areas can be tested.

Information obtained from paleomagnetic analyses of rocks regarding paleolatitudes and relative rotations of crustal blocks has helped to test tectonic concepts previously expressed on the basis of regional geology. Examples just in the vicinity of the Maya Terrane (Mexico, Gulf of Mexico, the Caribbean, and Central America) include the northward migration and rotation of the Baja California Peninsula [*Hausback,* 1984; *Hagstrum et al.,* 1985; *Beck,* 1986]; the large Mesozoic rotations [*Gose et al.,* 1982] as well as the tectonic transpression in the Sierra Madre Oriental [*Longoria,* 1985] in northeastern Mexico; the tectonic interpretation of the Trans-Mexican Volcanic Belt (TMVB) representing a zone of transtension being a developing boundary of the southern Mexico microplate [*Shurbet and Cebull,* 1984]; or the possible counterclockwise rotation of crustal blocks near the TMVB in central Mexico [*Urrutia-Fucugauchi and Bohnel,* 1987]. Farther south, a 28° counterclockwise rotation has been reported between the Oaxaca Terrane (late Paleozoic) and cratonic North America subsequent to the Oaxaca remagnetization [*McCabe et al.,* 1988]. The tectonics of the Maya Terrane relative to the formation of the Gulf of Mexico and the evolution of the nearby Caribbean and Central American regions have received considerable attention as well [e.g., *Pindell et al.,* 1988; *Ross and Scotese,* 1988; *Gose and Swartz,* 1977, *Gose,* 1983, 1985*a,b*].

In the paleogeographic reconstruction of continents, one of the most puzzling areas is that of nuclear Central America which includes the Maya Terrane and the Chortis Block which are referred to as northern or nuclear Central America [*Schuchert,* 1935]. The problem of its tectonic evolution is not only one of geometry and space but of the timing of events as well. *Carey* [1958] attempted to explain the origin of the Gulf of Mexico by pulling the Yucatan and Honduras blocks from within it, implying a counterclockwise rotation of 135° for the Yucatan Terrane and of 65° for the Honduras block, the postulated movement having occurred during the interval from Late Jurassic to Late Cretaceous times. The computer fit by *Bullard et al.* [1965] at the 500-fathom contour on the steep part of the continental edge shows a blank area between North and South America and leaves no space for pre-Mesozoic rocks known to exist in southern Mexico and northern Central America. Fortunately, there is a general consensus among present workers that the Gulf of Mexico Basin formed as a result of Mesozoic rifting between the North American plate and the Yucatan Terrane, as indicated by the presence of oceanic crust in the central gulf, by the lack of a subduction zone along any of the boundaries of this oceanic crust, and by the presence of extended continental crust peripheral to this oceanic crust [*Buffler et al.,* 1980; *Dickinson and Coney,* 1980; *Salvador and Green,* 1980; *Blair,* 1987].

From the few examples cited above, it is clear that the paleogeography of these regions needs to be better understood. Different models based on a variety of approaches and techniques have been proposed [e.g., *Van der Voo et al.,* 1976; *Ladd,* 1976; *Pindell and Dewey,* 1982; *Anderson and Schmidt,* 1983; *Pindell,*

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1985; *Salvador,* 1987; *Winker and Buffler,* 1988] and yet, major disagreements exist not only in regard to the past geometry of the area but also in relation to the timing of proposed displacements, varying from Late Triassic to post-Cretaceous time.

Paleomagnetic results now available from different areas around the Caribbean [*Watkins and Cambray,* 1970; *Harrison and Fink,* 1971; *MacDonald and Opdyke,* 1972; *Gose,* 1983, 1985*a,b; Gose and Testarmata,* 1983] and from Mexico [*Guerrero-Garcia,* 1975; *Gose and Sanchez-Barreda,* 1981; *Gose et al.,* 1982; *Urrutia-Fucugauchi,* 1984; *Urrutia-Fucugauchi et al.,* 1987] still do not provide an understanding of the tectonic evolution of the Yucatan Peninsula (Maya Terrane). The paleomagnetic data presented here assist by confirming the post-Jurassic tectonic stability of the Maya Terrane, and we believe that this places definitive constraints on the paleotectonics of the Maya Terrane and its relation to Caribbean tectonics.

GEOLOGIC SETTING

Nuclear Central America, as defined by *Schuchert* [1935] extends from southernmost Mexico (Chiapas) to northern Nicaragua. The basement rocks exposed in the area include pre-Permian metasedimentary and metavolcanic rocks underlying upper Paleozoic, Mesozoic and Cenozoic sedimentary strata [*Schuchert,* 1935; *Vinson and Brineman,* 1963; *Dengo,* 1969]. These rocks were affected by several episodes of igneous activity and tectonic deformation mainly during the middle Paleozoic, late Paleozoic and Late Cretaceous to early Tertiary. The major deformation events appear to be correlated with the Appalachian and Laramide orogenies of North America [*Dengo,* 1969; *Dengo and Bohnenberger,* 1969].

The Polochic-Motagua-Jocotan fault zone, cutting across central Guatemala and part of Honduras, divides the basement rocks in two areas of different metamorphic grade and, accordingly, *Dengo* [1969] defined two blocks on opposite sides of the fault zone (see Figure 1*a*). The Chortis block, south of the fault, is the area from southern Guatemala to northern Nicaragua in which the basement consists of phyllites and schists of the greenschist facies [*Zoppis Bracci,* 1957]. On the northern side, the Maya block includes northern Guatemala, Belize, and the Mexican states east of the Isthmus of Tehuantepec (see Figure 1*a*).

A sedimentary sequence unconformably overlies the basement and begins with Late Pennsylvanian-Early Permian clastic and carbonate rocks, folded and faulted during the Late Permian-Early Triassic [*Dengo and Bohnenberger,* 1969]. This sequence is overlain by Mesozoic-Cenozoic sediments in northern nuclear Central America which begins with the Todos Santos Formation, a transitional clastic phase, covered by a thick marine carbonate sequence. Deformed during the Laramide orogeny, these rocks constitute a series of parallel, northwest trending ranges that extend northeastward from the basement area to form the Yucatan platform, where they are covered by flat-lying, almost horizontal rocks [*Viniegra,* 1971].

STRATIGRAPHY

The Mesozoic sedimentary sequence in southern Mexico and western Guatemala begins with the terrigenous continental red beds of the Todos Santos Formation, originally named by *Sapper* [1894] after exposures near the village of the same name in western Guatemala. The formation is exposed in Chiapas [*Gutierrez-Gil,* 1956] and along the mountain front east of the Sierra de los Cuchumatanes as far east as Baja Verapaz, in Guatemala [*Clemons and Burkart,* 1971; *Anderson et al.,* 1973]. Conglomerate, sandstone, siltstone, and shale, with thin limestone beds locally present in the upper part, make up the sequence which ranges in thickness from more than 1200 m to less than 10 m [*Clemons and Burkart,* 1971; *Anderson et al.,* 1973]. Predeposition at uplift and faulting during the Triassic and part of Jurassic time [*Dengo,* 1969] produced a highly irregular topography responsible for the wide range of thickness.

Richards [1963] studied two sections of the Todos Santos Formation: (1) the Cerro Encantado section east of Cintalapa, Chiapas, and (2) the Ventosa section in the Altos Cuchumatanes in western Guatemala. At Ventosa, the formation is in fault contact with the Permian Chochal Formation, while at Cintalapa the basal conglomerate onlaps the Sierra Madre batholith. Richards redefined the Todos Santos Formation of Sapper into two different units: the Todos Santos (sensu strictu) composed of a lower conglomerate member and an upper siltstone shale member. The upper siltstone, shale, and limestone beds that form the upper part of the original Todos Santos of Sapper were named the San Ricardo Formation after a name informally used in Chiapas for those units [*Richards,* 1963; *Viniegra,* 1971; *Clemons et al.,* 1974].

At the Cerro Encantado section in Chiapas the San Ricardo Formation is divided into three units: a lower shale, a middle limestone referred to as the Palo Grande member, and an upper sandstone member. At the Ventosa section, Richards divided the San Ricardo into four units: a basal shale, a limestone referred to as the Ventosa member, a siltstone member, and a sandstone member known as the Rosario Sandstone (see Figure 2). The contact between the Todos Santos and San Ricardo Formation is conformable and gradational, being placed at the type section in Mexico where red beds grade upward into tan, green and gray shales [*Richards,* 1963].

On the basis of paleontological reports by *Burckhardt* [1930] and *Mullerried* [1936] of a Late Jurassic (Kimmeridgian) ammonite *Idoceras* sp. in the middle of the formation in Chiapas and of *Vinson's* (1962) findings of the Late Jurassic-Early Cretaceous foraminifera *Anchispirocyclina henbesti* Jordan and Applin in a basal conglomerate, Richards considers the age of the Todos Santos (sensu strictu) restricted to Late Jurassic. Although he does not assign the formation to any stage, other lines of evidence indicate that the unit is probably restricted to the Upper Oxfordian-Kimmeridgian stages [*Viniegra,* 1971]. The San Ricardo Formation is considered to be Late Jurassic-Early Cretaceous since the Ventosa limestone contains the same foraminifera found in the basal conglomerate [*Vinson,* 1962] whose age range is lower Kimmeridgian through lower Valanginian. *Mullerried* [1936] reported a Neocomian fauna in the upper part of the unit. A thick carbonate sequence beginning in late Neocomian overlies the San Ricardo Formation [*Vinson,* 1962; *Anderson et al.,* 1973; *Clemons et al.,* 1974]. The Todos-Santos-San Ricardo sequence thus ranges from early Late Jurassic to earliest Cretaceous (Neocomian) in age [e.g., *Richards,* 1963; *Castro Mora et al.,* 1975; *Alencaster,* 1977; *Clemons and Burkart,* 1971; *Anderson et al.,* 1973; *Blair,* 1981, 1987; *Michaud,* 1984]. According to *Blair* [1987] the Todos Santos and lower San Ricardo are coeval and interfinger in the subsurface with thick marine halite deposits of the Salina Formation.

During the field work for this study some samples were collected for palynological analysis, about 70 m above the limestone beds of the Palo Grande member, in the uppermost part of the red bed sequence near Cintalapa. The samples, analyzed at the University of Texas at Dallas, yielded a few specimens of *Cicatricosisporites* sp. which, although common in the Lower Cretaceous, has also been reported in Tithonian strata

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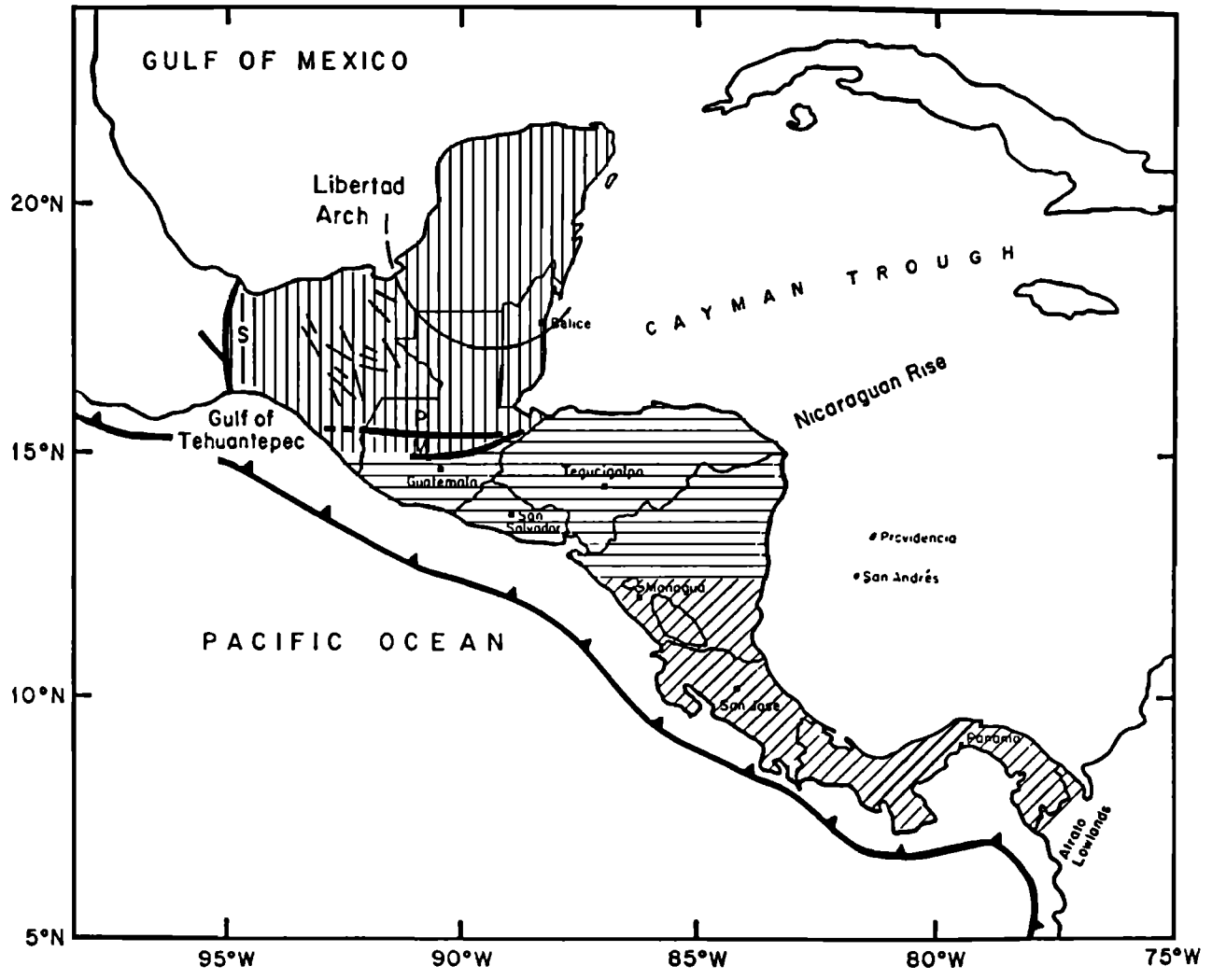


Fig. 1*a*. Map of nuclear Central America. The Maya Terrane, including the "Chiapas" Block, Chortis Block, and southern Central American Block as defined by *Dengo and Bohnenberger* [1969] are indicated by vertical, horizontal, and diagonal hachured patterns. The heavy lines show the Salina Cruz fault (S), La Libertad Arch, Polochic fault (P), Montagua fault (M), and the Middle America trench. (Modified from *Gose* [1985*a*,*b*].

[*Tschudy and Scott,* 1969; *Burger,* 1966]. The paleomagnetic samples were collected at this same locality and therefore are considered to be of upper Tithonian to Neocomian age.

Sampling and Treatment

A red bed sequence of hematitic shales and fine-grained sandstones is exposed a long a road cut between the villages of Francisco I. Madero and Lopez Mateos, about 10 km northwest of the town of Cintalapa, Chiapas, at 93.7°W, 16.8°N (Figure 1*b*). The red beds rest conformably upon limestone beds believed to be the Palo Grande member of *Richards* [1963] and structurally form the flank of a homocline that dips gently toward the northeast. The dirt road climbs up the face of the mountain north of the villages and exposes about 114 m of section, after which the tropical vegetation and thick soil cover renders sampling impossible.

One hundred and thirty-three oriented samples were collected in continuous stratigraphic succession throughout the sequence at intervals of about 80 cm. Strike and dip of beds were measured at several intervals so that appropiate structural correction could be applied. Upon return to the laboratory the cores were cut into cylindric specimens 2.54 cm long and the natural remanent magnetization (NRM) was measured using a ScT cryogenic magnetometer. The bottommost specimen of each core was used in the paleomagnetic analysis to minimize surface weathering effects.

Thermal demagnetization, performed on all samples at six progressive steps from 350°C up to 630°C, was accomplished in a horizontal, noninductively wound, 18-sample capacity furnace. All laboratory analyses were performed inside a mu-metal shielded room that houses the furnace and magnetometer and in which the maximum field is about 200 gammas. This minimized the acquisition of spurious thermal or viscous remanent magnetization components that might obscure the original magnetization vector.

The criteria for magnetic stability, as well as the range of the stable magnetization direction for each sample, was determined individually for each sample by analysis of the sample behavior in orthogonal vector demagnetization diagrams, stereographic projections and intensity plots.

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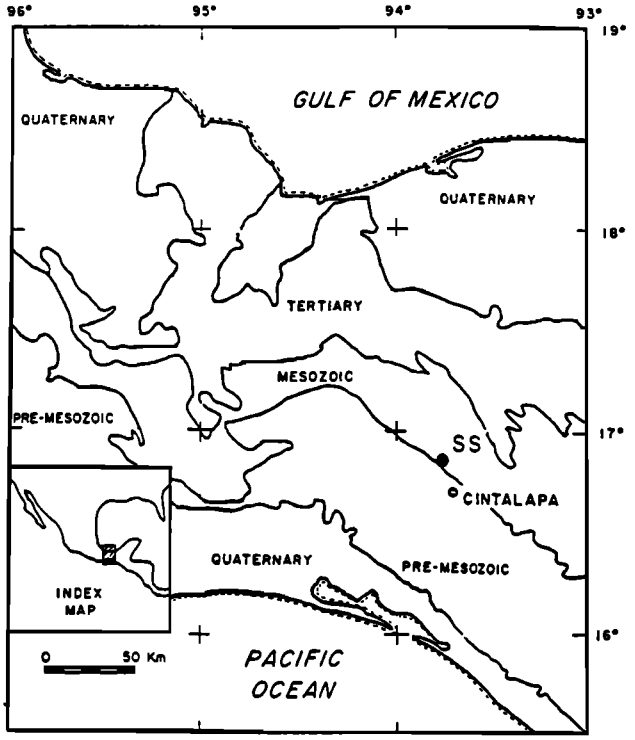


Fig. 1*b*. Location map showing a schematic generalized distribution of the geology [*Salas et al.*, 1968] in the Isthmus of Tehuantepec (see insert). The sampling site (SS) is shown by the solid dot northwest of Cintalapa.

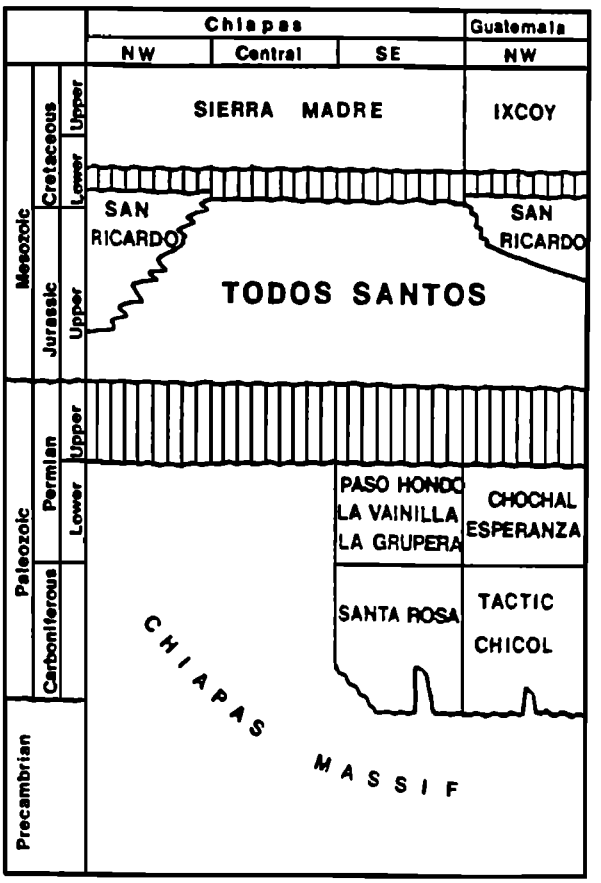


Fig. 2. Stratigraphic column for exposures in Chiapas, Mexico, and northwestern Guatemala [from *Blair*, 1987].

PALEOMAGNETIC RESULTS

The NRM declination, inclination and intensity of all 133 samples was measured on a SCT cryogenic magnetometer. Even from NRM directions, normal and reversed polarity were clearly evident throughout the sequence. The lower 60 m of normal polarity are followed by a short reversed interval of about 70 cm (two samples). Ten meters of normal polarity follow above, being overlaid by 45 m of a rather scattered reversed zone. This scatter is due to significant overprinting of the present-day field direction. The mean NRM declination of normal samples is 348° with an inclination of 22°, while the declination of reversed samples is 176° with an inclination of −1.5°. The two mean directions are approximately 160° apart. The mean NRM intensity is 0.644×10^{-2} A/m with a range from 0.066 to 2.45×10^{-2} A/m.

Since thermal demagnetization has proven very useful in analyzing the magnetization of red beds [e.g., *Dunlop and Stirling*, 1977; *Maslanyj and Collinson*, 1988] in which hematite is considered to be the carrier of part or all of the remanent magnetization, this method was applied in the present study.

Upon demagnetization several features were observed. While most of the normally magnetized samples showed small changes in declination and inclination, the upper reversed zone exhibited a better definition and the dispersion became smaller as demagnetization proceeded. Worth noting is the remarkable stability of the two samples that form the small reversed zone. Throughout the demagnetization procedure these two samples maintained a direction antipodal to the normal samples and which falls within the bulk of directions of the upper reversed interval.

The largest changes in directions of magnetization occur after heating to 350° and 500°C. Heating at 350°C resulted in a noticeable displacement of the normally magnetized samples of over 6° toward the northwest and a more pronounced clustering of the reversed samples, to a position almost antipodal to the samples with normal polarity.

Heating at 500°C further displaced the normal samples northwestwardly and the mean directions of normal and reversed samples are about 175° apart which indicates that most of the unwanted secondary magnetization has been removed and a reliable stable direction has been approached. The next steps up to 630°C merely caused changes of less than 3° both in mean declination and inclination. Some unstable samples were conspicuous throughout the demagnetization process. Their directions change drastically from one demagnetization step to the next, suggesting the presence of at least two components of magnetization with comparable blocking temperature.

Heating at 350°C resulted in a drop in the mean intensity of 30% in association with the changes in direction discussed above. Some reversed samples showed an initial increase in intensity followed by a decay, indicative of the removal of a soft component of magnetization of varying intensity aligned with the present day field direction. The normal samples exhibited continuous intensity decay throughout demagnetization. Typical changes in remanent intensity and directional behavior are shown in Figures 3*a* and 3*b*. As with the directions, the mean intensities of all samples stabilized after heating within the range from 500° to 580°C. At 630°C the remaining intensity was about 40% of the mean NRM intensity and yielded an average magnetization of 0.257×10^{-2} A/m. Considered separately, the natural

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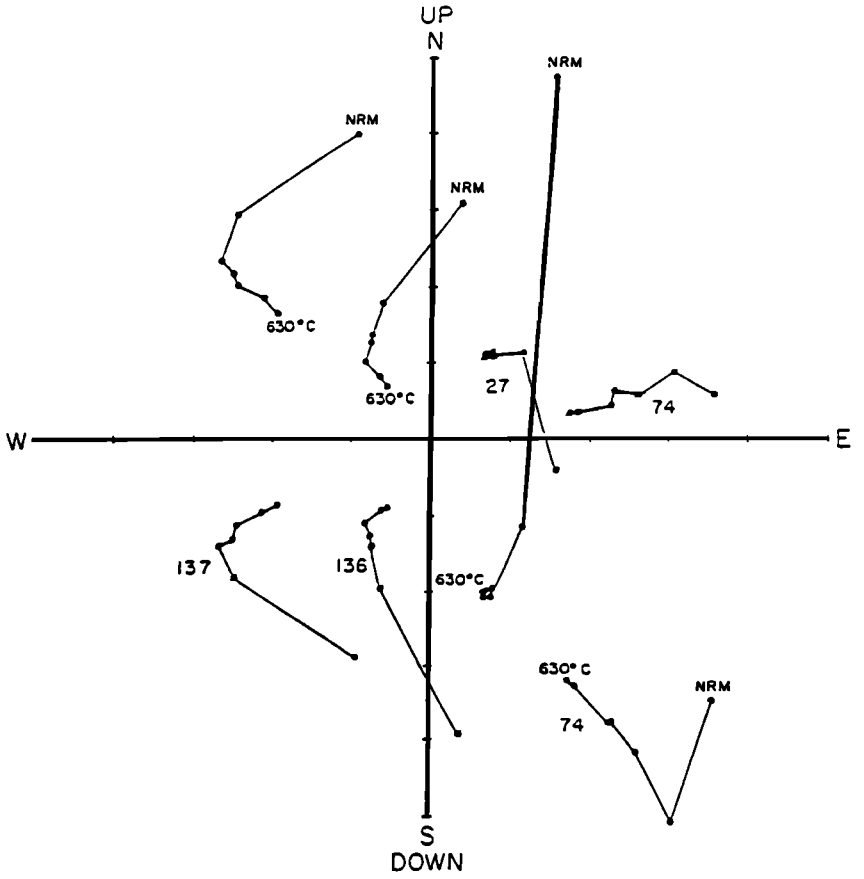


Fig. 3*a*. Examples of orthogonal vector plots for thermally demagnetized specimens showing directional stability with increasing temperature (see text for explanation). Horizontal projections are indicated by open symbols, with the NRM and 630°C steps lettered. Their corresponding vertical projections (either above or below the E-W line) are shown by solid symbols and by the sample number.

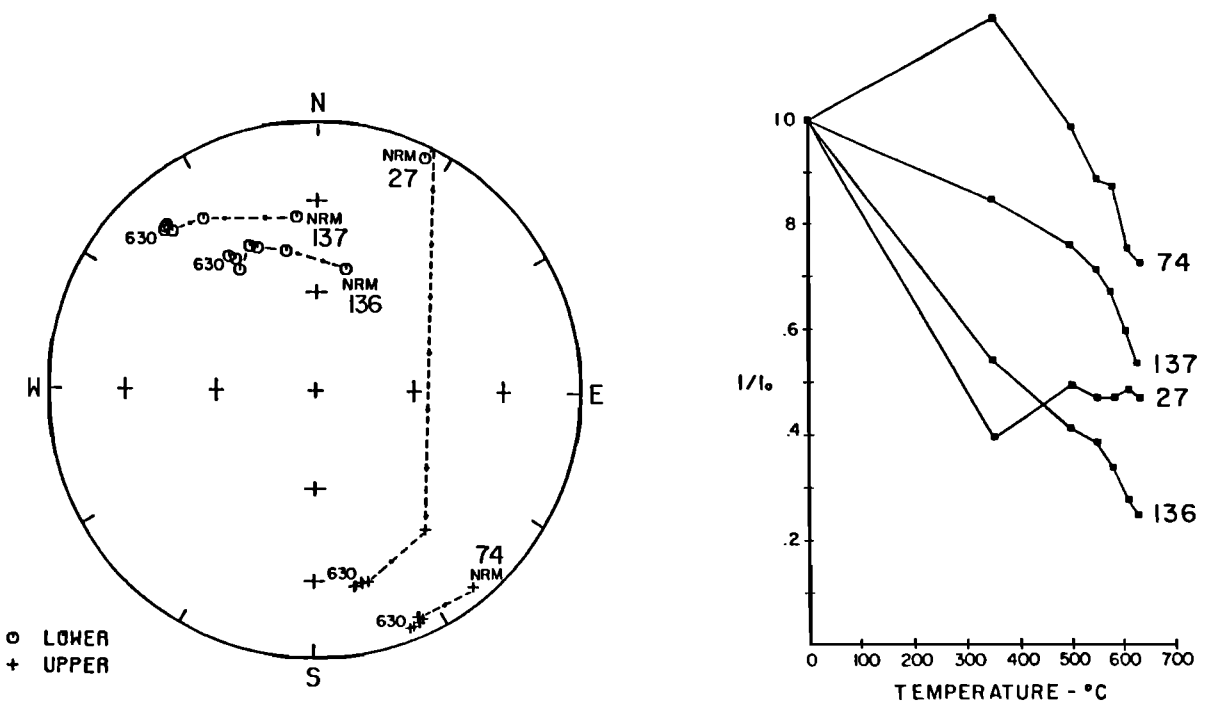


Fig. 3*b*. Equal-area projections and normalized remanent intensity versus temperature diagrams showing the behavior of some structurally corrected samples upon demagnetization. The path movement starts at NRM position and ends at 630°C position. Larger numbers indicate sample identification.

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remanent intensity of the normal samples was stronger by a factor of 2 than that of the reversed samples, probably reflecting the addition of a secondary component of normal polarity parallel to the present field. After demagnetization to 500°C, the remaining intensity of the normal and reversely magnetized samples was approximately equivalent suggesting that almost all of the secondary normal polarity overprint (essentially present-day) had been removed.

The response of the samples to the thermal treatment seems to indicate that the remanent magnetization is carried by at least two phases: a soft component aligned along the present field which is destroyed during heating to 350°C and which probably is due to the presence of goethite [Neel temperature, 120°C]. This iron oxyhydroxide is known to be the carrier of secondary components [*Strangway et al.*, 1968] and is commonly found in weathered rocks [*Strangway et al.*, 1968; *Strangway*, 1970; *Larson and Walker*, 1975]. The larger instability and scatter of surface samples as compared with subsurface samples has been generally attributed to surface weathering, with consequent formation of goethite [*Strangway et al.*, 1968; *Baag and Helsley*, 1974a]. Some hematite with a short relaxation time may also be associated with recent weathering and thus may be reponsible for a portion of the overprint.

The phase that carries the stable magnetization is made up of thermally distributed components and has properties similar to that described by several authors [*Irving and Opdyke*, 1965; *Herrero-Bervera and Helsley*, 1983]. Hematite with a wide blocking temperature spectrum appears to be responsible for the regular fall in intensity. Microscopic examination of a few thin sections show red pigmentation around the clastic grains as well as some black oxide grains. Heating above the Curie point of magnetite, 580°C, does not cause any appreciable change in direction or marked decrease in intensity. Thus the possibility of remanence being carried by magnetite appears to be ruled out.

The time of acquisition of magnetization in these red sediments cannot be fully determined from our data alone. The presence of a magnetic polarity stratigraphy supports an early remanence acquisition as argued by *Baag and Helsley* [1974b] as does the fact that the polarity of magnetization is not controlled by rock type which varies from coarse sandstones to well sorted siltstone.

STABILITY OF REMANENCE

Of primary importance to this study is the determination of a stable direction of magnetization that would yield a reliable pole position. The vector rotation criteria of *Zijderveld* [1967] was used for this purpose. The results of the progressive demagnetization steps for each specimen were analyzed from concurrent examination of orthogonal vector demagnetization diagrams and stereographic projections of the demagnetization data for each sample. The range of stable magnetization was selected from linear segments of the orthogonal vector diagrams and confirmed by examination of the stereographic directional data (see Figures 3*a* and 3*b*). Thirty-two samples that did not meet our stability requirements of at least two consecutive steps decreasing in intensity of the vector toward the origin were considered magnetically unstable and rejected. For the samples determined to have stable magnetization, least squares lines were fit to the directional vectors within the range of stability of each specimen. Where only two points were available the average was used to obtain the characteristic paleomagnetic direction of that specimen.

Upon completion of the demagnetization studies a further selection was made with the removal of twelve abnormal samples. These 12 samples, in spite of being determined as stable, plot in abnormal positions far from the mean normal or reversed groups (Figure 4). The cause of these abnormal directions is unknown but may reside in human errors such as mistaken orientation in the field, improper transfer of orientation marks in the laboratory, etc. Since these samples are not stratigraphically grouped, it is unlikely that they are representative of true magnetic field directions. They may, of course, represent intermediate

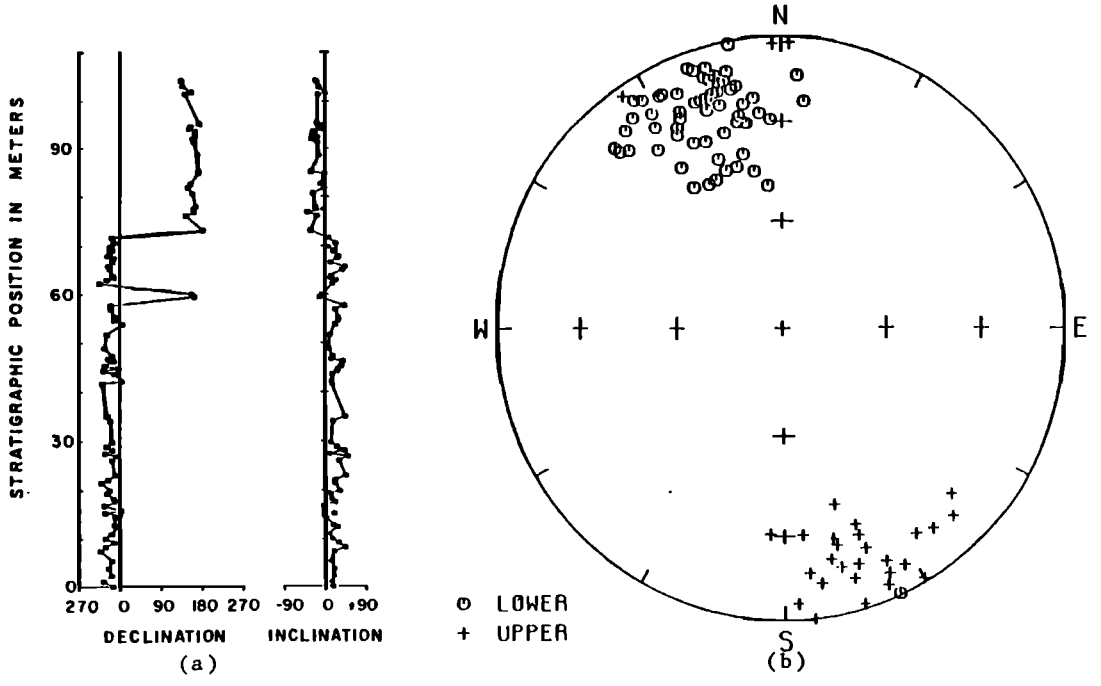


Fig. 4. Magnetization directions of final selected samples. (*a*) Stratigraphic plot showing the distribution of normal and reversed zones. (*b*) Equal-area plot of magnetization directions.

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directions of polarity transitions or excursions of the Earth's magnetic field. In any case, since the stable directions of these specimens is markedly different than that of the adjacent specimens, it seems justified to calculate the pole positions with and without these samples. A comparison of the statistical analyses of all the stable ($N=101$) samples with results from the selected group without the the abnormal samples ($N=89$) indicates that it is immaterial whether one includes or excludes the abnormal samples since the difference in both mean declination and inclination is less than one degree regardless of the selection criteria. The normal and reversed groups calculated separately are shown in Table 1. The results are plotted in Figure 4. With a total stable population of 89 specimens, a pole position was calculated at 160.0°E longitude, 69.8°N latitude, which compares favorably with the polar wander curve for North America for rocks of the same age.

DISCUSSION

At this point, it is important to emphasize the reliability of these red bed data. The samples studied in this paper do not show evidence of remagnetization events, as has been the case in most paleomagnetic studies in orogenic belts [e.g., *McCabe et al.*, 1988]. The samples analyzed in our present study do have definite polarity zones and antipodal reversed and normal directions (see Figure 4).

As summarized by *Steiner* [1983] and *May and Butler* [1986], eight reliable paleomagnetic poles from cratonic North America, mainly from the Colorado Plateau, cover the span of time from Sinemurian to late Tithonian and constitute the apparent polar wander (APW) curve for this geologic period against which our results can be compared. As suggested by *Beck* [1987], the APW path takes a right-angle bend at about 155 Ma. The results given in our paper certainly agree with the general APW for this interval for North America, and any small differences can easily be attributed to the rapid APW movement noted by M.E. Beck 1989 (personal communication, 1989).

The uppermost Jurassic Morrison Formation is believed to be Tithonian to late Kimmeridgian in age on the basis of magnetostratigraphic and paleontological data. The work on the Late Jurassic by *Steiner and Helsley* [1972, 1975], among others, indicates that up to Late Oxfordian time (Summerville Formation) the pole remained near to the Late Triassic pole. The overlying Morrison Formation records about 20° of apparent motion between late Oxfordian-early Kimmeridgian and 10° during Kimmeridgian-Tithonian, along a line of latitude (present-day coordinates) between published Triassic and Cretaceous poles.

The pole position obtained in this study is in excellent agreement with the pole determined for the upper part of the Morrison Formation which is thought to be of similar age to the San Ricardo Formation (Figure 5) uncorrected for the 3.9° to 10° clockwise rotation of the Colorado Plateau [*Bryan and Gordon*, 1986; *Steiner*, 1986]. Both cones of confidence include each other's result. When calculated separately by polarities, the results suffer no major change and the cones, while enlarged, would still include the upper Morrison pole. The possibility that the San Ricardo may be older than Kimmeridgian might allow a counterclockwise rotation of about 20° from the Sumerville pole position to the Morrison pole position. The presence of *Cicatricosisporites* sp., however, supports a maximum age of Tithonian, that is, equivalent to upper Morrison. If the age of the sampled sequence is indeed early Neocomian, then the results should lie upon the fringe of Cretaceous poles. Granted that it may be somewhat fortuituous within the resolution of the paleomagnetic methods, the agreement with the results of the upper Morrison Formation is remarkable, especially in view of the fact that the San Ricardo sampling area lies 2000 km south from the Colorado Plateau. This agreement itself may be indicative that indeed the age of the San Ricardo is comparable to that of the Morrison, at least in the area of this work. The postulated rotation of the Colorado Plateau because of its nearness to the pole produces comparable uncertainties, and this is not of significance for this discussion.

The mobilistic solutions proposed to accommodate the Yucatan Peninsula demand that either the peninsula be extracted

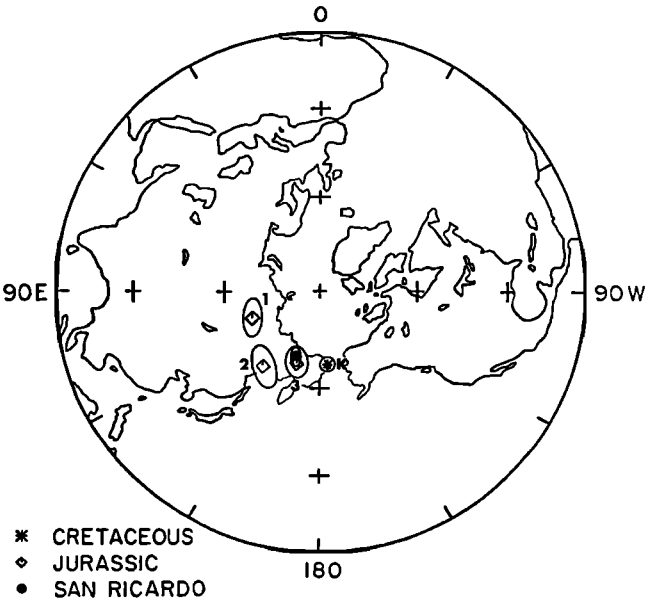


Fig. 5. San Ricardo pole position shown with the background of published Cretaceous poles and Jurassic poles cited in text: 1, Summerville Formation; 2 and 3, Lower and Upper Morrison Formation. The ovals of confidence of the San Ricardo and Upper Morrison poles are plotted for comparison.

TABLE 1. Analyses of Magnetization Directions of Selected Data

	D, deg	*I*, deg	*N*	*R*	*K*	α_{95}	Longitude, °E	Latitude, °N	*dp*	*dm*
First selection	340.4	20.4	101	22.48	11.7	4.3	160.5	70.0	2.4	4.5
Second selection	340.3	19.9	89	85.53	25.4	3.0	160.0	69.8	1.7	3.2
Selected Normal	340.2	21.5	62	59.52	24.7	3.7	162.4	70.0	2.1	3.9
Selected Reversed	340.5	16.4	27	26.07	28.1	5.1	164.8	69.2	2.8	5.5

D, mean declination with reversed samples inverted to normal polarity; *I*, mean inclination with reversed samples inverted to normal polarity; *N*, number of samples; *R*, resultant vector; *K*, precision parameter; α_{95}, semiangle of 95% confidence cone; *dp*, *dm*, semiaxis of error ellipse.

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from within the present Gulf of Mexico or that the peninsula was a southward directed continuation of the North American continent with later rotations resulting in its present position. Other tectonic models suggest a previous westerly position along the present-day Pacific coast of Mexico. Inasmuch as the paleomagnetic method provides control regarding paleolatitude and orientation of crustal blocks but yields no information about longitudinal displacements, each issue is discussed separately.

The fact that the observed pole position for the San Ricardo Formation corresponds to the age equivalent pole for North America, seems to preclude any larger rotation of the type proposed by *Carey* [1958], *MacDonald and Opdyke* [1972], *Walper and Rowett* [1972], and *Uchupi* [1973] since the earliest Cretaceous. Any of these post-Jurassic rotations require a paleomagnetic pole for the San Ricardo Formation far removed from the observed positions of the Upper Jurassic-earliest Cretaceous poles. Clearly, that is not observed. Correction for the 3.9° rotation of the Colorado Plateau displaces the upper Morrison pole leftward a few degrees, and separates the San Ricardo and upper Morrison poles by an apparent post-Jurassic clockwise rotation of the Yucatan Peninsula which is bound to be smaller than 3.9°, considering that the sampling site is farther south from the Colorado Plateau. Such a small rotation falls within the uncertainty of the paleomagnetic method.

The possibility of a drastic lateral translation of the whole Maya Terrane along a line of paleolatitude with respect to the San Ricardo pole, so as to fit northwestern South America into the Gulf of Mexico, cannot be detected with paleomagnetism. This is very unlikely since such a displacement would probably also result in a detectable difference with respect to this North American pole. Moreover, large post-Jurassic longitudinal motions cannot be explained without disrupting the Mesozoic sedimentary sequence. This sequence begins at least in Late Jurassic time and is exposed along the Gulf Coast in the Geologic Map of Mexico [*Salas et al.*, 1968] and extends into northern Central America [*Dengo*, 1969; *Dengo and Bohnenberger*, 1969]. Facies maps show no major break and overlapping models of the type proposed by *Bullard et al.* [1965] must take this geologic evidence into account, in the timing of proposed events. The same line of reasoning applies to north-south displacements of the Maya block. Again, the close correspondence with the pole of the age-equivalent uppermost Morrison Formation from cratonic North America and the geologic evidence mentioned above rules out large post-Jurassic latitudinal movements. Finally, it seems unlikely that this area is a displaced terrane whose final resting place fortuitously lies just at the right latitude that the declination would also be fortuitously just right to match the North American Lower Cretaceous position.

Certainly, the area under study has undergone deformation and is still tectonically active. Nevertheless, the stable directions of magnetization of the samples taken for this paleomagnetic study of the San Ricardo Formation indicate that no discernible rotations or latitudinal displacements of the Maya block have occurred since at least Neocomian (Early Cretaceous) time. This result suggests that the Maya Terrane was already part of the North American plate by this time. It is important to mention that to assume tectonic continuity between Chiapas and the Yucatan Peninsula may not be valid as pointed out by *Gose and Sanchez-Barreda* [1981] and reported also by *Urrutia-Fucugauchi et al.* [1987], since major discontinuities such as La Libertad Arch (see Figure 1*a*), the fault system of northern Chiapas, and the faults in northern Belize may represent effective tectonic boundaries [*Viniegra*, 1971]. A better and more positive case ought to be made for the tectonic significance of these data. If the data are correct, and they certainly appear to be, then either (1) the assembly of Middle America was substantially complete by latest Jurassic time, or (2) the "Maya Terrane" moved into place along a line of latitude, with just the right amount of rotation about a vertical axis. Of course, a very unlikely third possibility exists: that the "Maya Terrane" had a very complicated history with several periods of rotation and that the agreement between its pole and the North American reference poles is just a coincidence. Thus the fact that the earliest Cretaceous pole from the San Ricardo Formation red beds agrees well with the corresponding segment of the North American APW path suggests strongly that the "Chiapas" Block was essentially fixed with respect to cratonic North America by Neocomian times as reported by other workers [*Gose and Sanchez-Barreda*, 1981; *Gose*, 1985*a*,*b*; *Salvador*, 1987].

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